

Section 2.1 - Question 3

Which of the intervals $(0, 5), (0, 5], [0, 5), [0, 5], (1, 4], [2, 3], (2, 3)$ contains:

- a) 0** $[0, 5), [0, 5]$
- b) 1** $(0, 5), (0, 5], [0, 5), [0, 5]$
- c) 2** $(0, 5), (0, 5], [0, 5), [0, 5], (1, 4], [2, 3]$
- d) 3** $(0, 5), (0, 5], [0, 5), [0, 5], (1, 4], [2, 3]$
- e) 4** $(0, 5), (0, 5], [0, 5), [0, 5], (1, 4]$
- f) 5** $(0, 5], [0, 5]$

Section 2.1 - Question 5

For each of these pairs of sets, determine whether the first is a subset of the second, the second is a subset of the first, or neither is a subset of the other.

a)

the set of airline flights from New York to New Delhi, the set of nonstop airline flights from New York to New Delhi

The second set is a subset of the first set.

b)

the set of people who speak English, the set of people who speak Chinese

Neither is a subset of the other.

c)

the set of flying squirrels, the set of living creatures that can fly

The first set is a subset of the second set.

Section 2.1 - Question 7

Determine whether each of these pairs of sets are equal.

a)

$\{1, 3, 3, 3, 5, 5, 5, 5, 5\}, \{5, 3, 1\}$

Equal

b)

$\{\{1\}\}, \{1, \{1\}\}$

Not equal

c)

$\emptyset, \{\emptyset\}$

Not equal

Section 2.1 - Question 9

For each of the following sets, determine whether 2 is an element of that set.

a)

$\{x \in \mathbb{R} \mid x \text{ is an integer greater than } 1\}$

$2 \in \{x \in \mathbb{R} \mid x \text{ is an integer greater than } 1\}$

b)

$\{x \in \mathbb{R} \mid x \text{ is the square of an integer}\}$

$2 \notin \{x \in \mathbb{R} \mid x \text{ is the square of an integer}\}$

c)

$\{2, \{2\}\}$

$2 \in \{2, \{2\}\}$

d)

$\{\{2\}, \{\{2\}\}\}$

$2 \notin \{\{2\}, \{\{2\}\}\}$

e)

$\{\{2\}, \{2, \{2\}\}\}$

$2 \notin \{\{2\}, \{2, \{2\}\}\}$

f)

$\{\{\{2\}\}\}$

$2 \notin \{\{\{2\}\}\}$

Section 2.1 - Question 11

Determine whether each of these statements is true or false.

a)

$$0 \in \emptyset$$

false

b)

$$\emptyset \in \{0\}$$

false

c)

$$\{0\} \subset \emptyset$$

false

d)

$$\emptyset \subset \{0\}$$

true

e)

$$\{0\} \in \{0\}$$

false

f)

$$\{0\} \subset \{0\}$$

false

g)

$$\{\emptyset\} \subseteq \{\emptyset\}$$

true

Section 2.1 - Question 13

Determine whether each of these statements is true or false.

a)

$$x \in \{x\}$$

true

b)

$$\{x\} \subset \{x\}$$

false

c)

$$\{x\} \in \{x\}$$

false

d)

$$\{x\} \in \{\{x\}\}$$

true

e)

$$\emptyset \subseteq \{x\}$$

true

f)

$$\emptyset \in \{x\}$$

false

Section 2.1 - Question 18

Use a Venn diagram to illustrate the relationships $A \subset B$ and $A \subset C$.

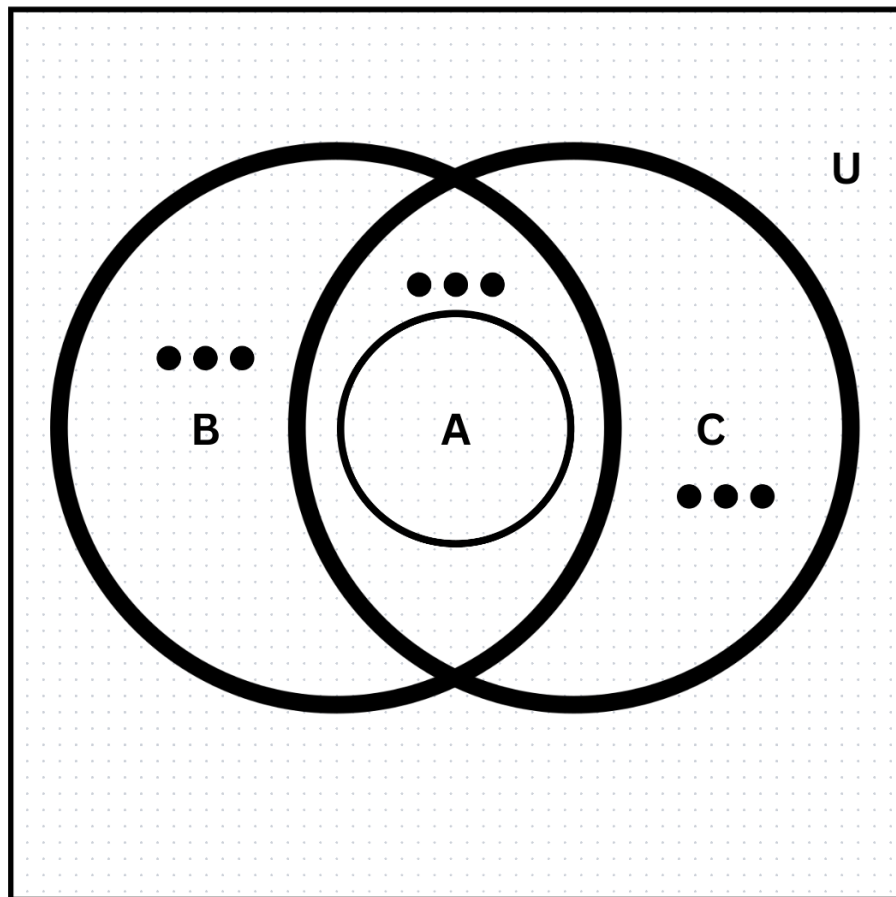


Figure 1: alt text

The dots indicate that the regions are not empty

Section 2.1 - Question 23

Find the power set of each of these sets, where a and b are distinct elements.

a)

$\{a\}$

$\{\emptyset, \{a\}\}$

b)

$\{a, b\}$

$\{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

c)

$\{\emptyset, \{\emptyset\}\}$

$\{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$

Section 2.1 - Question 25

How many elements does each of these sets have where a and b are distinct elements?

a)

$$P(\{a, b, \{a, b\}\})$$

8

b)

$$P(\{\emptyset, a, \{a\}, \{\{a\}\}\})$$

16

c)

$$P(P(\emptyset))$$

- $P(x)$ represents the power set of x
- The power set of $\emptyset$ is $\{\emptyset\}$
- The power set of $\{\emptyset\}$ is $\{\emptyset, \{\emptyset\}\}$
- So $P(P(\emptyset)) = \{\emptyset, \{\emptyset\}\}$ which has 2 elements

Section 2.1 - Question 29

Let $A = \{a, b, c, d\}$ and $B = \{y, z\}$. Find

a)

$$A \times B.$$

$\{(a, y), (a, z), (b, y), (b, z), (c, y), (c, z), (d, y), (d, z)\}$

b)

$$B \times A.$$

$\{(y, a), (y, b), (y, c), (y, d), (z, a), (z, b), (z, c), (z, d)\}$

Section 2.1 - Question 35

Find A^2 if

a)

$$A = \{0, 1, 3\}.$$

$$A^2 = A \times A = \{(0, 0), (0, 1), (0, 3), (1, 0), (1, 1), (1, 3), (3, 0), (3, 1), (3, 3)\}$$

b)

$$A = \{1, 2, a, b\}.$$

$$A^2 = A \times A = \{(1, 1), (1, 2), (1, a), (1, b), (2, 1), (2, 2), (2, a), (2, b), (a, 1), (a, 2), (a, a), (a, b), (b, 1), (b, 2), (b, a), (b, b)\}$$

Section 2.1 - Question 37

How many different elements does $A \times B$ have if A has m elements and B has n elements?

$A \times B$ has $m * n$ elements

Section 2.1 - Question 44

Prove or disprove that if A , B , and C are nonempty sets and $A \times B = A \times C$, then $B = C$.

Hypothesis: A , B , and C are nonempty sets and $A \times B = A \times C$

Conclusion: $B = C$

Prove by contradiction:

1. Assume hypothesis is true - A , B , and C are nonempty sets and $A \times B = A \times C$
2. Assume conclusion is false - $B \neq C$
3. If $B \neq C$ and B is a nonempty set, then there exists an element b in B that is not in C
4. Since $A \times B = A \times C$, then for every element a in A , (a, b) is in $A \times B$ if and only if (a, b) is in $A \times C$
5. Since (a, b) is in $A \times B$, then (a, b) is in $A \times C$
6. Since (a, b) is in $A \times C$, then b is in C
7. This contradicts the assumption that b is not in C
8. Therefore, $B = C$ necessarily by contradiction

Section 2.1 - Question 47

Find the truth set of each of these predicates where the domain is the set of integers.

a)

$$P(x): x^2 < 3$$

$\{-1, 0, 1\}$

b)

$$Q(x): x^2 > x$$

$\{\dots, -3, -2, 2, 3, \dots\}$

c)

$$R(x): 2x + 1 = 0$$

$\{-1\}$

Section 2.1 - Question 48

Find the truth set of each of these predicates where the domain is the set of integers.

a)

$$P(x): x^3 \geq 1$$

$\{1, 2, 3, \dots\}$

b)

$$Q(x): x^2 = 2$$

$\emptyset$

c)

$$R(x): x < x^2$$

$\{\dots, -2, -1, 2, 3, \dots\}$

Section 2.2 - Question 3

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{0, 3, 6\}$. Find

a)

$$A \cup B.$$

$\{0, 1, 2, 3, 4, 5, 6\}$

b)

$$A \cap B.$$

$\{3\}$

c)

$$A - B.$$

$\{1, 2, 4, 5\}$

d)

$$B - A.$$

$\{0, 6\}$

Section 2.2 - Question 7

Prove the domination laws in Table 1 by showing that

a)

$$A \cup U = U.$$

1. $A \cup U = U$
2. $A \cup U = A \cup (A' \cup A)$
3. $A \cup U = (A \cup A') \cup A$
4. $A \cup U = U \cup A$
5. $A \cup U = U$
6. Therefore, $A \cup U = U$

b)

$$A \cap \emptyset = \emptyset.$$

1. $A \cap \emptyset = \emptyset$
2. $A \cap \emptyset = A \cap (A' \cap A)$
3. $A \cap \emptyset = (A \cap A') \cap A$
4. $A \cap \emptyset = \emptyset \cap A$
5. $A \cap \emptyset = \emptyset$
6. Therefore, $A \cap \emptyset = \emptyset$

Section 2.2 - Question 9

Prove the complement laws in Table 1 by showing that

a)

$$A \cup !A = U.$$

1. $A \cup !A = U$
2. $A \cup !A = A \cup (A' \cup A)$
3. $A \cup !A = (A \cup A') \cup A$
4. $A \cup !A = U \cup A$
5. $A \cup !A = U$
6. Therefore, $A \cup !A = U$

b)

$$A \cap !A = \emptyset.$$

1. $A \cap !A = \emptyset$
2. $A \cap !A = A \cap (A' \cap A)$
3. $A \cap !A = (A \cap A') \cap A$
4. $A \cap !A = \emptyset \cap A$
5. $A \cap !A = \emptyset$
6. Therefore, $A \cap !A = \emptyset$

Section 2.2 - Question 25

Prove the first distributive law from Table 1 by showing that if A , B , and C are sets, then $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

1. $A \cup (B \cap C) = \{x \mid x \in A \text{ or } x \in B \text{ and } x \in C\}$
2. $= \{x \mid x \in A \text{ or } (x \in B \text{ and } x \in C)\}$
3. $= \{x \mid (x \in A \text{ or } x \in B) \text{ and } (x \in A \text{ or } x \in C)\}$
4. $= (A \cup B) \cap (A \cup C)$
5. Therefore, $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Section 2.2 - Question 27

Let $A = \{0, 2, 4, 6, 8, 10\}$, $B = \{0, 1, 2, 3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8, 9, 10\}$. Find

a)

$$A \cap B \cap C.$$

$\{4, 6\}$

b)

$$A \cup B \cup C.$$

$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

c)

$$(A \cup B) \cap C.$$

$\{4, 5, 6, 8, 10\}$

d)

$$(A \cap B) \cup C.$$

$\{0, 2, 4, 5, 6, 7, 8, 9, 10\}$

Section 2.2 - Question 29c

Draw the Venn diagrams for $(A \cap !B) \cup (A \cap !C)$

Simplify:

1. $(A \cap !B) \cup (A \cap !C)$
2. $A \cap (!B \cup !C)$
3. $A \cap !(B \cap C)$
4. This is the intersection of A with the area of B that is not in C and the area of C that is not in B

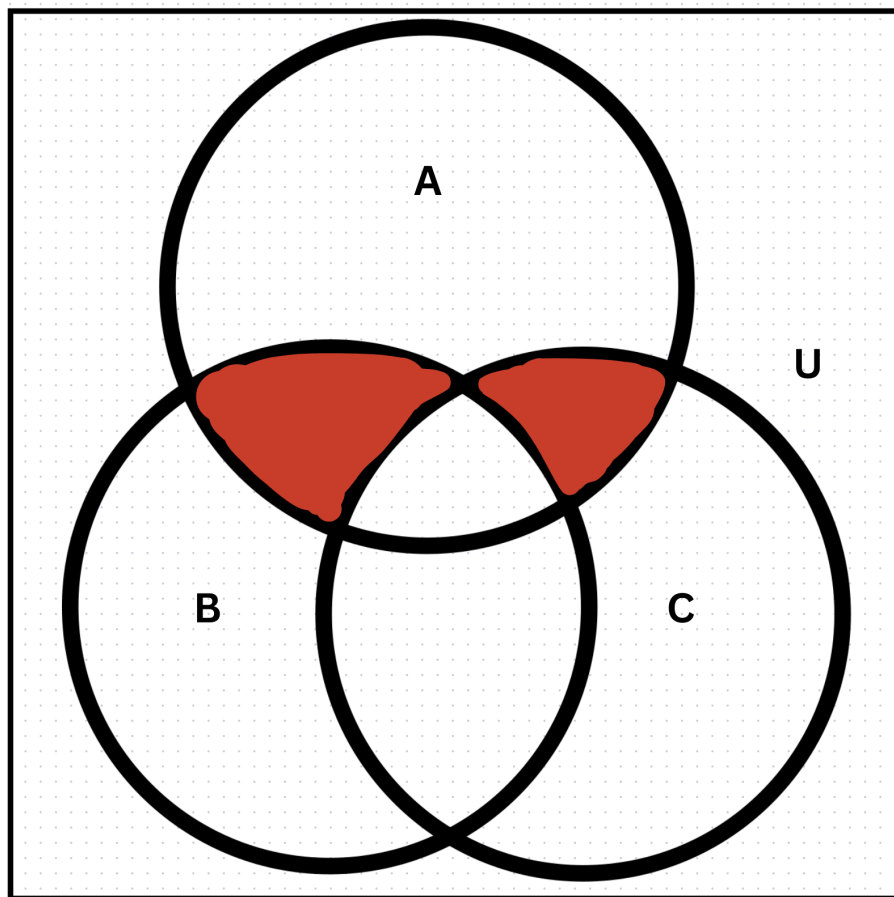


Figure 1: alt text

Section 2.2 - Question 31

What can you say about the sets A and B if we know that

a)

$$A \cup B = A?$$

- B does not have any elements that are not in A.
- B is a subset of A

b)

$$A \cap B = A?$$

- All of the elements of A are in B.
- A is a subset of B

c)

$$A - B = A?$$

- None of the elements of B are in A.
- $A \cap B = \emptyset$

d)

$$A \cap B = B \cap A?$$

- Nothing

e)

$$A - B = B - A?$$

- A and B are equal or have no elements in common

Section 2.2 - Question 39

Find the symmetric difference of the set of computer science majors at a school and the set of mathematics majors at this school.

$\{ x \mid x \text{ is a computer science major} \oplus x \text{ is a mathematics major} \}$

Section 2.2 - Question 43d

Show that if A is a subset of a universal set U , then $A \oplus !A = U$.

1. $A \oplus !A = U$
2. $A \oplus !A = (A - !A) \cup (!A - A)$
3. $A \oplus !A = A \cup !A$
4. $A \oplus !A = U$
5. Therefore, $A \oplus !A = U$

Section 2.2 - Question 47

Suppose that A, B, and C are sets such that $A \oplus C = B \oplus C$.
Must it be the case that $A = B$?

1. $A \oplus C = B \oplus C$
2. $A \oplus C = (A - C) \cup (C - A)$
3. $B \oplus C = (B - C) \cup (C - B)$
4. $(A - C) \cup (C - A) = (B - C) \cup (C - B)$
5. Yes

Section 2.2 - Question 49

If A, B, C, and D are sets, does it follow that $(A \oplus B) \oplus (C \oplus D) = (A \oplus D) \oplus (B \oplus C)$?

1. $(A \oplus B) \oplus (C \oplus D) = (A \oplus D) \oplus (B \oplus C)$
2. $(A \oplus B) \oplus (C \oplus D) = ((A - B) \cup (B - A)) \oplus ((C - D) \cup (D - C)) = (A - B - C - D) \cup (B - A - C - D) \cup (C - D - A - B) \cup (D - C - A - B)$

Yes

Section 2.2 - Question 50

Show that if A and B are finite sets, then $A \cup B$ is a finite set.

Direct Proof:

1. A and B are finite sets
2. Therefore, A and B have a countable number of elements
3. So we can suppose that A has n elements and B has m elements, where n and m are integers
4. Then $A \cup B$ has $n + m$ elements
5. Because integers are closed under addition, $n + m$ is an integer
6. Therefore, $A \cup B$ has an integer number of elements and is a finite set

Section 2.2 - Question 54

Let $A_i = \{\dots, -2, -1, 0, 1, \dots, i\}$. Find

a)

The union of n intervals, where each interval is denoted as A_i for $i = 1, 2, 3, \dots, n$.

$\{\dots, -2, -1, 0, 1, \dots, n\}$

b)

The intersection of n intervals, where each interval is denoted as A_i for $i = 1, 2, 3, \dots, n$.

$\{\dots, -2, -1, 0, 1\}$

Section 2.2 - Question 56

Find the Union and Intersection of n intervals, where each interval is denoted as A_i for $i = 1, 2, 3, \dots$

a)

$$A_i = \{i, i + 1, i + 2, \dots\}.$$

- Union $\{1, 2, 3, \dots\}$
- Intersection $\emptyset$

b)

$$A_i = \{0, i\}.$$

- Union $\{0, 1, 2, \dots\}$
- Intersection $\{0\}$

c)

$$A_i = (0, i), \text{ that is, the set of real numbers } x \text{ with } 0 < x < i.$$

- Union $\{x \in \mathbb{R} \mid x > 0\}$
- Intersection $\{x \in \mathbb{R} \mid 0 < x < 1\}$

d)

$$A_i = (i, \infty), \text{ that is, the set of real numbers } x \text{ with } x > i.$$

- Union $\{x \in \mathbb{R} \mid x > 1\}$
- Intersection $\emptyset$

Section 2.2 - Question 59

Using the same universal set as in the last exercise, find the set specified by each of these bit strings.

a)

11 1100 1111

{1, 2, 3, 4, 7, 8, 9, 10}

b)

01 0111 1000

{2, 4, 5, 6, 7}

c)

10 0000 0001

{1, 10}

Section 2.2 - Question 67

Let A and B be the multisets $\{3 \cdot a, 2 \cdot b, 1 \cdot c\}$ and $\{2 \cdot a, 3 \cdot b, 4 \cdot d\}$, respectively. Find

a)

$$A \cup B.$$

$$\{3 \cdot a, 3 \cdot b, 2 \cdot a, 2 \cdot b, 1 \cdot c, 4 \cdot d\}$$

b)

$$A \cap B.$$

$$\{2 \cdot a, 2 \cdot b\}$$

c)

$$A - B.$$

$$\{3 \cdot a, 1 \cdot c\}$$

d)

$$B - A.$$

$$\{3 \cdot b, 4 \cdot d\}$$

e)

$$A + B.$$

$$\{5 \cdot a, 5 \cdot b, 1 \cdot c, 4 \cdot d\}$$